

Hariprakash Karthikeyan

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WORK EXPERIENCE

Butler/Till

Rochester, New York

Data Scientist

Aug 2025 – Dec 2025

- **Led development and ownership** of hierarchical Bayesian Media Mix Models across ~170 DMAs, replacing heuristic planning and supporting data-driven budget reallocation with improved ROI visibility.
- **Implemented scalable MMM pipelines** incorporating adstock carryover and Hill-type saturation effects, reducing manual scenario analysis time by ~40–50% and enabling rapid evaluation of channel and geo-level budget tradeoffs.
- **Evaluated model performance** using rolling historical holdout validation, achieving **RMSE = 0.84 versus baseline assumptions**, ensuring stable and reliable geo-level estimates across all markets.

Buffalo Solar

Buffalo, New York

Data Science Intern

Jun 2025 – Jul 2025

- **Developed predictive time-series models** to identify anomalous energy production patterns across **468 solar sites**, replacing manual monitoring and enabling operations teams to prioritize inspections and maintenance more effectively.
- **Deployed end-to-end modeling and analytics pipelines** into a production **Streamlit** application used by engineering and operations teams for continuous site-level monitoring and decision support.
- **Refined anomaly detection logic** using historical residual analysis and threshold-based validation, achieving ~90% alignment across 468 sites and improving monitoring reliability while reducing false positives and manual review effort.

University of Rochester Medical Center

Rochester, New York

Research Assistant

Feb 2025 – May 2025

- **Applied statistical and machine learning techniques** to longitudinal EHR data spanning **2,000+ patients**, replacing manual chart review and enabling scalable cohort-level clinical analysis and downstream predictive modeling.
- **Developed NLP pipelines** to extract structured phenotypes from unstructured clinical notes, producing a **clean, analysis-ready cohort of 845 patients** from fragmented and noisy EHR data.
- **Designed clinically interpretable, time-aligned data workflows** validated by physicians and reused across multiple research analyses, reducing manual abstraction effort and accelerating study turnaround.

PROJECTS

End-to-End Distributed Sentiment Analytics (Databricks, Spark)

- **Implemented distributed Spark pipelines** on Databricks to process and analyze sentiment across 100k+ text records, enabling scalable aggregation and comparative analysis across partitions and time windows.
- **Tracked model performance and experimentation** using MLflow, accelerating iteration speed by ~40% and preventing regressions during sentiment model refinement.
- **Integrated Python (PySpark, Pandas), MLflow, and Delta Lake** to support reproducible experimentation and production-ready analytics workflows suitable for large-scale product monitoring.

Urban Mobility & Demand Analytics (NYC Taxi Data)

- **Applied time-series analysis** to large NYC taxi data (~60GB) using **Python, Pandas, NumPy, and statsmodels**, uncovering demand seasonality, peak-hour dynamics, and spatial patterns to inform demand forecasting and operational planning.
- **Engineered temporal and spatial features** (hour-of-day, day-of-week, zone-level aggregations) using **SQL and PySpark**, transforming raw trip logs into analysis-ready datasets suitable for scalable forecasting and downstream modeling.

SKILLS

Programming & Querying: Python, SQL, R

Machine Learning & Statistics: Statistical Modeling, Predictive Modeling, Supervised Learning, Time-Series Forecasting, Experimentation & A/B Testing

ML Frameworks & Libraries: scikit-learn, PyTorch, TensorFlow

NLP & LLMs: NLP Pipelines, Text Feature Extraction, Embeddings, Retrieval-Augmented Generation (RAG)

Data Systems: Spark, Databricks, Relational Databases

MLOps & Deployment: ML Pipelines, Model Validation & Deployment, MLflow

Visualization & Tools: Plotly, Matplotlib, Git, Linux

Cloud & Process: AWS, Agile / Scrum

EDUCATION

University of Rochester

Rochester, New York

MS, Data Science

Aug 2024 – Dec 2025

- **Relevant Coursework:** Data Science at Scale, Statistical Machine Learning, Network Science Analytics, Database Systems

SRM Institute of Science and Technology

Chennai, India

Bachelor of Technology, Electronics and Communication Engineering

Aug 2020 – May 2024